A prospective assay reveals scaffold-driven hypothesis-space expansion

Jing-Rung Huang[1,*] and Wen-Hsiang Lu[1]

[1] Department of Computer Science and Information Engineering, National Cheng Kung University, Tainan 701, Taiwan

[*] Corresponding author: Jing-Rung Huang (p78084063@mail.ncku.edu.tw)
ORCID: Jing-Rung Huang, 0000-0003-4776-3550; Wen-Hsiang Lu, 0009-0002-5149-6790

Article | manuscript for scientific discussion

Abstract

Scientific discovery can require changing not only a hypothesis but the language in which hypotheses are expressed. We introduce a prospective, replayable assay for bounded hypothesis-space expansion: candidates must leave a frozen language, fit observations, commit to an intervention before outcome reveal, outperform the observation-optimal incumbent comparator and survive held-out falsification. Externally typed proposals succeeded in 142 of 400 synthetic worlds. Deterministic compositional search paired with a frozen, family-aligned motif-to-basis realizer succeeded in all 400 known-family worlds and 100 worlds from one search-side holdout. Model-free replay reproduced every verdict because deterministic components supplied the evaluated representations, fitted expressions and interventions. A grammar-constrained model proposer produced validated edits in 15 of 96 worlds but did not outperform random composition. The assay therefore measures system-level representational escape while showing that causal attribution requires field-level provenance across model and scaffold components.

Scientific change can alter the representational vocabulary in which hypotheses are stated[1,2,3]. Computational creativity and constructive induction distinguish exploration within a space from transformations of that space[4,5,6,7,8,9]. Symbolic regression, program search and language-model-guided systems discover equations, algorithms and hypotheses[10,11,12,13,14,15,16,17], while multi-agent systems automate broader workflows[18,19,20].

Studies of language-model creativity reach mixed conclusions: some systems exceed population-average scores on divergent-thinking tasks, yet stronger human responses remain competitive, and judged creative outputs do not establish scientific representational change[21]. A recent position paper describes the generation of new explanatory premises as an abductive "jump" that current language models may lack[22]. Neither divergent-thinking scores nor that historical argument establishes whether a system left a specified hypothesis language, survived a prospective intervention, or which model-scaffold component supplied the move.

Recent work now explicitly targets expanding principle or model spaces. PiEvo treats discovery as optimization over an evolving principle space across four benchmarks[23]. Model Discovery Agent couples a language-model proposer to Bayesian experiment design in an open model setting spanning physics, chemistry and biology[24]. HypoArena evaluates prospective hypothesis discovery in 988 cases across six domains and 15 frontier models[25]. EvoSCM evolves structural causal models through intervention and prospective prediction[26]. Our contribution is not the first system to expand a principle or model space. It is an assay that requires canonical non-membership relative to a frozen incumbent language, commitment-before-outcome-reveal prediction, exact replay and component-level attribution of the resulting theory.

Related diagnostic benchmarks study underdetermined hypothesis generation and inspiration-based task decomposition[27,28]. A reconstructed molecular-genetics task found mainly incremental assistance from generative AI, but its historical case study does not establish a general incapacity[29]. Our separation of observational fit from intervention and falsification follows the interventionist distinction between association and causal prediction[30,31].

Here we operationalize bounded hypothesis-space expansion. We freeze an incumbent representational language, generate executable candidates and require a six-gate verdict: incumbent adequacy (J0), canonical structural non-membership (J1), candidate adequacy (J2), a discriminating committed prediction (J3), prospective intervention gain (J4) and survival on separate held-out falsification cases (J5; Fig. 1). Here, escape denotes structural non-membership in a frozen representation language together with prospective improvement over a selected observational comparator; it does not establish functional inexpressibility within the entire incumbent language. Candidate prose, model confidence, embedding distance and semantic novelty do not enter the verdict.

Our experiments validate the assay and reveal a component-attribution limit. External typed proposals and generic compositional search cross the frozen language reliably, whereas fixed-space alternatives do not. However, code-path ablation shows that the successful search condition's representations, fitted equations, ranking and interventions were deterministic; removing model output leaves all 500 jump and 300 control verdicts unchanged. The audit demonstrates how a system-level discovery benchmark can misattribute successful explanatory content to a model when deterministic scaffolding supplies the decisive representation, fit and intervention. The contribution is therefore the prospective assay, the typed search result and a method for auditing causal system attribution-not evidence that a language model produced the successful escapes.

Results

A prospective criterion for hypothesis-space expansion

Figure 1 | A prospective assay for hypothesis-space expansion

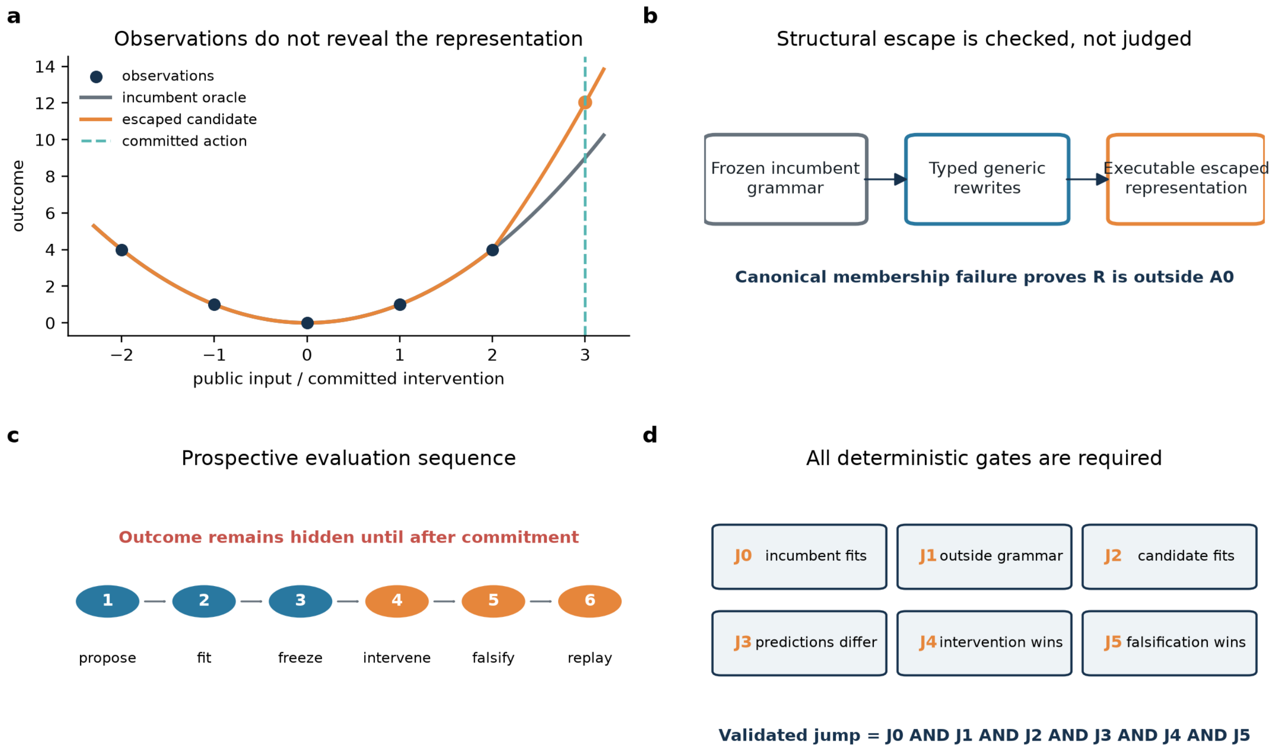


Figure 1 | A prospective assay for hypothesis-space expansion. Panel a illustrates observational equivalence and a committed action; panels b-d are schematics of structural escape, commitment-before-outcome-reveal ordering and the conjunctive J0-J5 gates. JSR denotes the proportion of worlds with at least one candidate passing every gate.

Each synthetic world contains observations compatible with an incumbent model, a frozen typed-graph language, hidden intervention outcomes and separate held-out falsification cases. The comparator minimizes observation loss over the frozen finite incumbent program set, with canonical-JSON tie-breaking; it is not optimized on intervention outcomes. Before any intervention outcome is revealed, a candidate representation, executable expression, selected action, prediction and split hash are committed. A validated jump is the conjunction $J0 \wedge J1 \wedge J2 \wedge J3 \wedge J4 \wedge J5$ (Fig. 1). This design turns "leaving a hypothesis space" into a replayable event rather than a judgement about wording or apparent novelty.

Typed proposals and fixed-language structural controls

Figure 2 | Typed proposals and their gate attrition

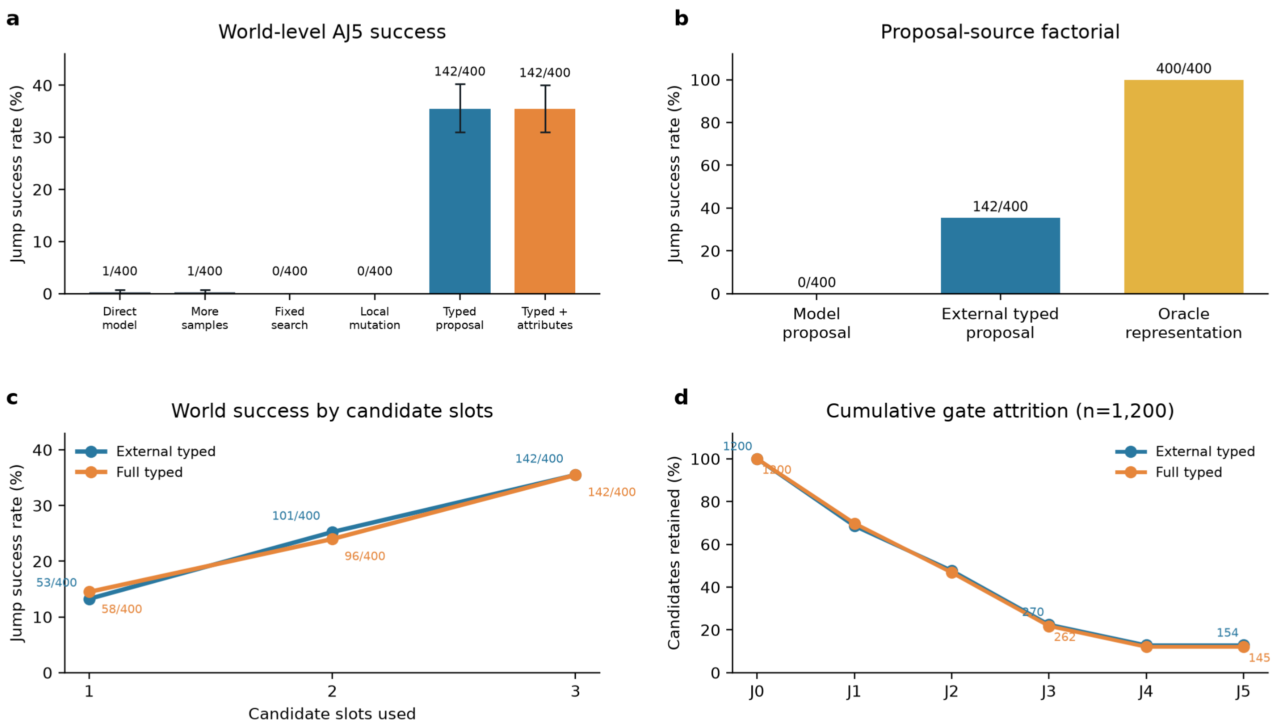


Figure 2 | External typed proposals and their gate attrition. Panels a and b report world-level jump success rate (JSR) for n=400 worlds per condition; intervals are prospectively specified family-stratified bootstrap 95% intervals. Panel c reports cumulative successful-world counts as one, two and three slots are admitted. Panel d reports cumulative candidate retention from J0 to J5 for 1,200 candidates per condition; candidates are not independent replicates. Internal condition codes are retained in Methods and source data. All AJ5 conditions recorded 0/200 control-world false jumps.

In AJ5, direct (B0) and sampling-matched (B1) model proposals each succeeded in 1/400 worlds; fixed-space reasoning (B2), attribute-only mutation (B3) and value-only mutation succeeded in none. External typed proposals (B4 and B5) each succeeded in 142/400 (35.5%; Fig. 2a). All eight prospectively specified contrasts with B0-B3 had positive paired differences. The archived bootstrap tail quantities are retained in the source data but are not treated as null-calibrated P values (Supplementary S10).

Holding the downstream path fixed, the proposal-source comparison found 0/400 successes from model-proposed representations, 142/400 from external typed proposals and 400/400 from oracle representations (Fig. 2b). The system can execute a supplied changed representation, but this does not establish model necessity after deterministic fitting because the second call received a fitted expression and maximum-separation intervention that were programmatically enforced.

Gate attrition clarifies the result (Fig. 2d). From 1,200 jump-world candidates each, B4 retained 823 at J1, 573 at J2, 270 at J3 and 154 at J4-J5; B5 retained 838, 562, 262 and 145. B0 retained 65, 22, 4 and 1, while B1 retained 546, 375, 118 and 1, showing that these executable model proposals usually failed after rather than before structural evaluation. Controls still produced 112 and 110 B4/B5 candidates through J3, but none passed J4: prospective outcomes, not J0-J3 alone, eliminated these apparent escapes.

Fixed-language conditions serve as structural negative controls: because they cannot satisfy J1 by design, their zero jump rates do not estimate a general limitation of search or reasoning.

Generic rewrites and fixed motif realization produce validated representations

Figure 3 | Generic search with fixed motif realization

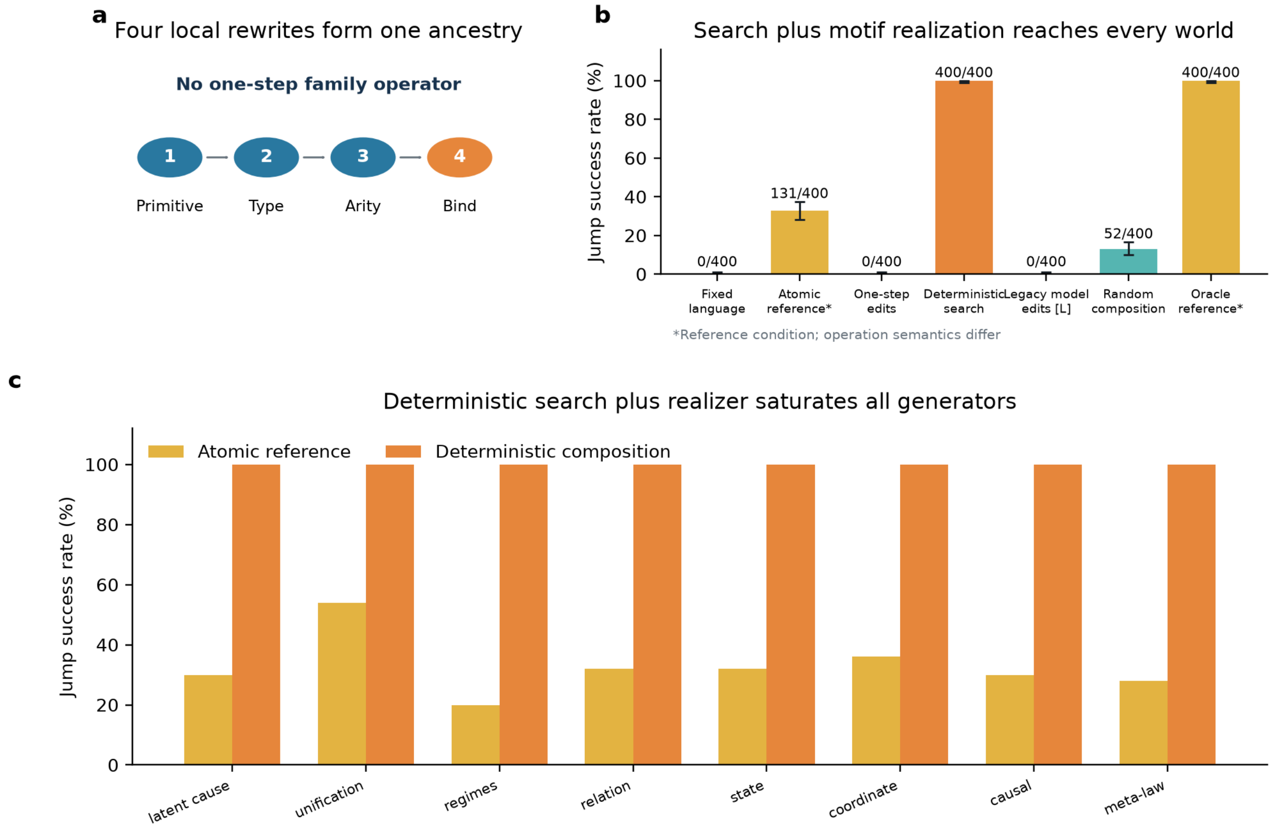


Figure 3 | Generic search with fixed motif realization. Panel a is schematic. Panel b reports known-family JSR with Wilson 95% intervals for n=400 worlds per condition. Atomic and oracle conditions are references with different operation semantics. [L] The legacy model interface failed before executable proposal evaluation; its zero does not isolate conceptual proposal ability. C3 and random composition compare complete search-and-selection policies. Panel c reports descriptive rates for 50 worlds per family. Deterministic composition combines generic edit search with a fixed, family-aligned motif-to-basis realizer; saturation is within-generator reliability, not independent family replication.

CJ5 replaced nine high-level mutations with 29 local graph and syntax-tree rewrites. Node addition, type, observability, arity and argument binding were separate steps; no primitive accepted a family label, target distance or outcome. C3 traversed 48 four-step branches. At each edit, a deterministic realizer mapped graph motifs to nine fixed basis libraries and fitted coefficients from public observations. The resulting fits and public-action prediction separations entered the outcome-blind scores; diversity-aware final ranking then retained three terminal candidates before commitment and outcome reveal. Those hand-authored bases aligned with the procedural family mechanisms, including the held-out triadic-product basis; C3 therefore tests search over motif triggers, not invention of an unseen functional form.

Across eight known families (n=400 worlds), fixed-space search (C0) and depth-one alternatives (C2) succeeded in none. The atomic reference (C1) succeeded in 131 (32.75%), random four-step paths (C_rand) in 52 (13.0%) and C3 in all 400 (Fig. 3). C3 exceeded C_rand by 0.87 (95% CI 0.845-0.895; adjusted P=3.00x10-4). Its family-wise saturation demonstrates within-generator reliability and assay separability, not eight independent replications or a frontier benchmark.

The C3-C_rand contrast compares complete search-and-selection policies: C3 ranks using observational compatibility and prospective discrimination, whereas C_rand selects by seed-fixed structural hash. It does not isolate proposal generation from candidate ranking.

Code-path ablation attributes C3 success to deterministic scaffolding

Inspection of the frozen C3 path showed that representation, expression, ranking and intervention selection occurred deterministically before the model response. Response fields for representation, expression and intervention were overwritten; only an explanation remained, and explanations do not enter J0-J5. We therefore replaced both model calls with a valid empty explanation and recomputed every theory, commitment and verdict without inference.

All 2,400 candidate verdicts matched the archive. The replay retained the pooled 400 known-family and 100 held-out jump-world successes and the structurally guaranteed 0/300 control false jumps. Phi-4 output was therefore not semantically necessary for C3. The perfect result belongs to the typed deterministic search, hand-authored motif-to-basis realizer and evaluation scaffold. More generally, the audit shows why scientific-discovery evaluations need code-path attribution: system success alone does not identify which component supplied the explanatory content.

Interface controls separate serialization from executable proposal

Historical model-planned composition scored 0/400, but offline reconstruction showed that this could not isolate proposal quality. All 1,200 responses were non-empty and JSON-extractable, yet none contained the required outer plan list; every response reached its completion cap. Contract inspection also showed that the prompt omitted executor argument keys. On a fixed outcome-blind panel of 96 historical worlds, increasing Phi-4 precision or budget, substituting DeepSeek, or allowing one validator repair left every legacy-interface condition at 0/96. Native DeepSeek returned reasoning but no separate answer to the parser. These failures were dominated by pre-executable attrition and do not establish conceptual incapacity.

A separately frozen grammar-constrained DeepSeek condition supplied exact parser syntax but no truth, fitted law, intervention outcome or semantic repair. All 4,608 plan opportunities were schema-valid and 3,939 were dynamically executable. The resulting model-proposed graph edits yielded 15/96 successful worlds, concentrated in meta-law and unification, compared with 16/96 for outcome-blind random composition on the identical panel (Fig. 4). The deterministic scaffold still supplied motif realization, fitting, ranking and intervention. The redesigned two-stage, grammar-constrained interface yielded executable model-generated edit plans, but did not show that the model independently supplied the realized law or outperformed random primitive search. Full precision, budget, repair, token-cap and supplied-representation diagnostics are reported in Supplementary Sections S12-S16 and Extended Data Figures 1-3.

Counterfactual replay tests predictive content and field binding

Figure 4 | Component attribution of system-level escape

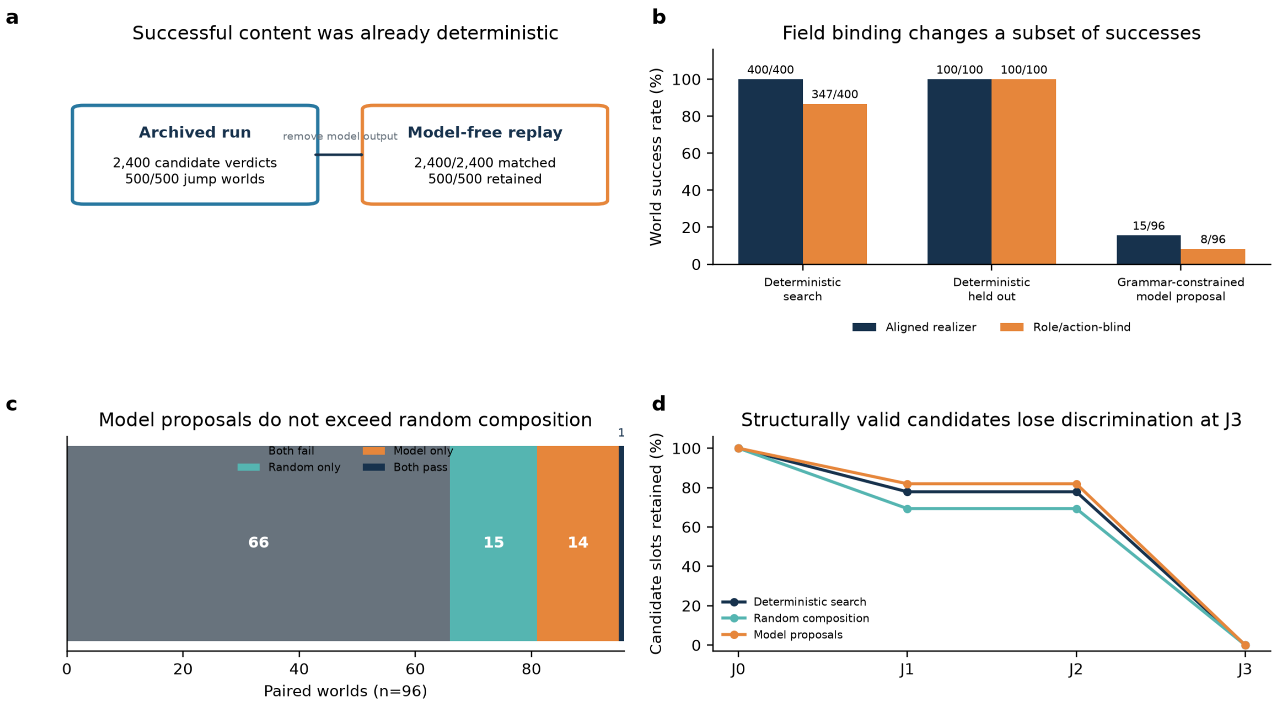


Figure 4 | Component attribution of system-level escape. Panel a compares archived deterministic-search verdicts with inference-free replay; all 2,400 candidate verdicts matched. Incumbent substitution forces zero prediction separation and J3 failure by construction; it does not establish that the basis library is irreplaceable. Panel b compares aligned and role/action-blind binding for C3 on 400 known-family and 100 held-out worlds, and grammar-constrained model proposals on the fixed 96-world panel. Panel c pairs the model proposer with random composition on those same 96 worlds; similar aggregate success does not establish policy equivalence. Panel d reports cumulative candidate-slot attrition under motif-disabled replay; candidate slots are not independent replicates. The post-confirmatory counterfactual made zero model calls and fixed archived candidates rather than rerunning search.

We froze an inference-free realizer-dependence audit after the preceding component findings and before counterfactual replay (Fig. 4). It retained every archived candidate representation and slot from C3 and C_rand across 400 known-family and 100 held-out worlds, plus the grammar-constrained DeepSeek candidates on the 96-world panel. The aligned policy reproduced all archived gates and verdicts without mismatch. We then replaced every non-incumbent motif realization with the exact incumbent expression while retaining the outside-space candidate graph and deterministically recommitting its maximum-separation public action. C3 fell from 400/400 and 100/100 to 0/400 and 0/100; C_rand fell from 52/400 and 13/100 to zero; DeepSeek fell from 15/96 to zero. Despite this, 1,168/1,500 C3, 1,040/1,500 C_rand and 236/288 DeepSeek candidate slots still passed cumulative J1-J2; none passed J3. Because this intervention sets the candidate expression equal to the incumbent expression, prediction separation is zero and failure at J3 follows by construction. This structural negative control verifies removal of non-incumbent predictive content; it does not establish that this particular basis library is the only realization mechanism capable of supporting escape.

A second policy retained each detected motif's algebraic term shape but rebound variables canonically by public field type and lexical order, without node roles or inspection of which fields changed across intervention queries. It retained 347/400 known-family and 100/100 held-out C3 successes, 8/96 DeepSeek successes, and 57/400 plus 13/100 C_rand successes. C3 losses were confined to hidden regimes (26/50 retained) and meta-law (21/50); DeepSeek retained 5/12 meta-law and 3/12 unification worlds. This policy is not semantics-free because it preserves the motif's algebra. Within this fixed pipeline, changing field binding affects a subset of archived successes even when motif algebra is retained. The incumbent-substitution control does not compare alternative realizers capable of producing non-incumbent expressions.

Leave-one-signature-out replay sharpened the attribution. Masking relation_arity_3 removed all 100 held-out C3 successes. Masking multi_argument_function removed 50 C3 worlds and all 15 DeepSeek worlds. Masking unobserved_dependency removed 100 C3 worlds; the remaining active signatures each accounted for 50, except shared_rule_binding, whose removal caused no world loss because alternative multi_argument_function candidates covered the same unification worlds. These are post-confirmatory fixed-candidate sensitivities, not reruns of search under a new realizer.

Transfer to one held-out structural family

The second study reserved triadic_relation_reification from development, pilot inference and known-family confirmation. C3 succeeded in 100/100 held-out worlds, C_rand in 13/100, and C0, C1, C2 and C_self in none (difference from C_rand 0.87, 95% CI 0.80-0.93; adjusted P=3.00x10-4).

The holdout is limited: an earlier family used binary property-to-relation change, the generic language already included reification and the shared realizer already contained the triadic-product basis. The arity-three target structure, generator and instances were new, but the realization mechanism was not. This is therefore a search-side structural-family holdout; one adjacent family cannot support broad external generalization.

A worked prospective escape

Figure 5 | One complete prospective escape

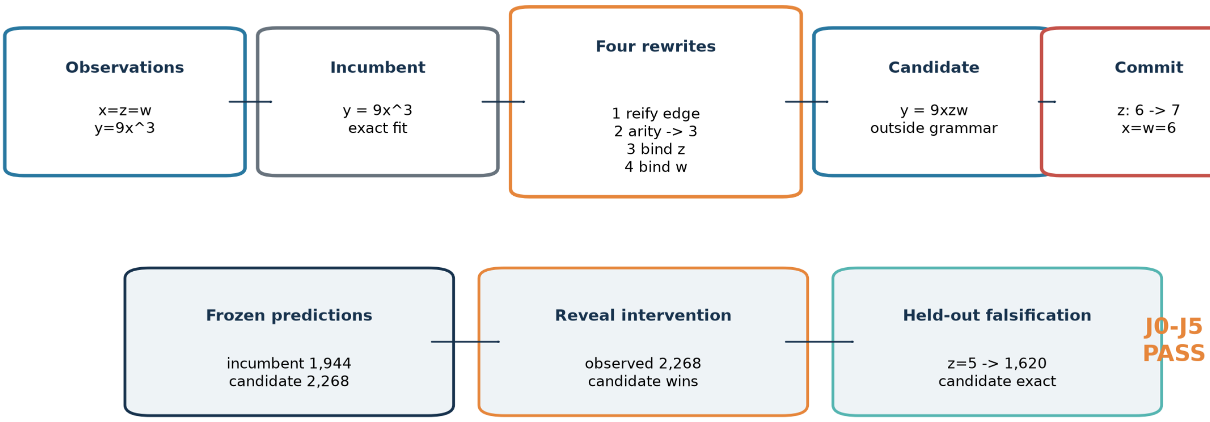


Figure 5 | Worked held-out example. Correlated observations make the cubic incumbent and triadic candidate observationally identical. A committed intervention separates them before outcome reveal, and a separate held-out case falsifies the incumbent.

In one held-out world, observations with x=z=w gave $y=9x^3$, so the incumbent $y=9x^3$ and candidate y=9xzw both fit exactly. Four generic edits reified the incumbent prediction edge, changed the new relation to arity three, and bound z and w; the pre-specified realizer supplied the xzw basis and fitted its coefficient (Fig. 5). Before outcome reveal, the assay committed to setting z from 6 to 7 at x=w=6. The incumbent predicted 1,944; the candidate predicted 2,268; the revealed outcome was 2,268. A separate held-out falsification case at z=5 yielded 1,620, again exact for the candidate. The representation failed the frozen language on relation count, type, arity bindings and result structure, and passed J0-J5.

Specificity and replay

The zero false-jump count requires qualification. In C3 controls, 800 of 900 candidates passed J1, 567 passed J2 and 283 passed J3, but none passed J4. The null was therefore non-trivial through the prospective commitment gate. An offline check verified that the selected observation-optimal comparator exactly matched the simulator on all 1,625 intervention and falsification cases across the 300 CJ5 control worlds (and all 1,125 cases in the 200 AJ5 controls). Under this verified condition, no candidate can achieve the strictly positive error reduction required by J4 and J5. The observed 0/300 is best interpreted as a structural-specificity check under the exact simulator, not an empirical false-positive estimate for noisy science; the Wilson upper bound is descriptive only.

All 10,800 AJ5 and 16,800 CJ5 selected candidates replayed exactly, including 35,533 CJ5 ancestry records. No confirmatory world was excluded and no shard was rerun.

Discussion

This work contributes an assay for a specific event: an executable candidate leaves a frozen hypothesis language and pays for that move with a prospective prediction that beats the selected observation-optimal incumbent and survives separate held-out falsification cases. Canonical non-membership, commitment-before-outcome-reveal ordering and exact replay distinguish this event from judged novelty or retrospective fit.

The code-path and counterfactual audits are methodological results. They demonstrate that a system-level discovery benchmark can assign successful explanatory content to a language model even when deterministic scaffolding supplies the decisive representation, basis, fit, ranking and intervention. Substituting incumbent expressions eliminated prospective discrimination by construction; role/action-blind rebinding supplies the non-trivial sensitivity result. Conversely, legacy model-planned zero rates were dominated by pre-executable attrition, whereas executable AJ5 model proposals usually failed at later gates. A grammar-constrained interface restored executability and exposed model-generated graph topology, but the deterministic scaffold still supplied realization and evaluation, all successes used one motif and aggregate performance did not exceed random composition. End-to-end failure can therefore underestimate what reaches the evaluator, just as end-to-end success can overattribute explanatory content to a model. Claims of model-driven representation change require field-level provenance throughout the committed theory.

A system-level escape claim, a model-contribution claim and a representation-invention claim require different evidence. The first concerns whether the committed executable theory passes the assay. The second requires tracing model-authored fields into the evaluated artifact and testing whether those fields affect the outcome. The third additionally requires showing that the decisive explanatory forms were not supplied by a benchmark-aligned realization library. Passing the first test does not establish the other two.

External validity is the principal limitation. The worlds are synthetic and noiseless; C3 is saturated; search strata, generators and motif-to-basis realizer were co-designed; and the operator language embeds graph, function, relation and binding priors. The original confirmation used one model; the extension sampled only one additional checkpoint on a fixed subset, and its native condition was not compute-matched. Only one conceptually adjacent, search-side family was held out. World seeds quantify within-generator reliability. The higher-level generalization units are nine structural families, only one of which was held out, so small world-level P values do not substitute for structural breadth.

A decisive extension should freeze the current assay and add independently authored families, distractor primitives, deeper compositions, stochastic observations, partial observability and at least one recognizable mechanistic simulation. Broader open and proprietary model comparisons could then use the tested separation of reasoning and grammar-valid answer stages, while reporting dynamic executability and J1-J5 rather than treating format compliance as scientific success. These experiments are proposed, not part of the present evidence.

The conclusion is narrow but durable: hypothesis-space expansion can be measured prospectively and replayed exactly; typed deterministic search paired with a hand-authored algebraic realizer aligned with the procedural generators can produce it in controlled worlds; and causal component auditing can reveal when apparent capability or incapacity instead belongs to the scaffold or interface.

Methods

Study design and protocol freezing

AJ5 and CJ5 protocols were frozen in Git before their reported confirmatory model calls. AJ5 uses commit 895ebb9118ffd0046825b88868621f2a70f69f61. CJ5 uses 65f20874e16bddf8a7ae36996395ff52b27153b7, with a documented correction at 7ecb977 and held-out unlock at 27ee542. The commits are publicly retrievable but unsigned, and no independent registry or transparency-log timestamp was located. We therefore describe the studies as prospectively specified and commit-frozen, not formally preregistered. Pilot seeds were excluded; no confirmatory result changed families, operators, thresholds, budgets or seeds.

Model and inference

Both studies used frozen microsoft/phi-4 revision 2db69c1c3e91a05d2c64a3185acfbaf36f744e25, vLLM 0.10.2, dynamic bitsandbytes 4-bit quantization, a 4,096-token context, temperature 0.2, top-p 0.95 and completion cap 700. AJ5 sampling proposals used temperature 0.7. Each request had a deterministic seed. No fine-tuning or cross-world adaptation occurred.

Minimal targeted sensitivity extension

The original Phi-4 artifacts were preserved at commit ae1ede683fdef09f2bf60f6e1052b60394ad6cf8, tag nmi-phi4-frozen-2026-09 and archive branch nmi-phi4-frozen-archive-2026-09. A separate extension protocol, panel, model configurations and generation code were frozen at commit 320eb29b33ddaed596b0fe7b3f1d5895c706f311 and amended only for operational execution at f846c89287e379fe313551c47765c24f2abf4959. New outputs used the nmi_minimal_sensitivity_v1 namespace and could not overwrite historical files.

An outcome-blind SHA-256 ranking selected 12 of the 50 historical CJ5 seeds before any new model call: 30014, 30012, 30029, 30025, 30011, 30023, 30000, 30032, 30001, 30037, 30002 and 30015. Applying the same seeds to all eight known families produced a fixed 96-world paired panel. The positive control used the first five selected seeds in each family (n=40). This extension is a targeted sensitivity analysis, not a replacement confirmatory population.

The frozen run matrix contained Phi-4 8-bit C_self on 96 worlds; DeepSeek matched C_self with reasoning_effort=none on the same worlds; DeepSeek native C_self with reasoning_effort=max; and a supplied-correct-representation DeepSeek positive control. The DeepSeek service identified the checkpoint as deepseek-ai/DeepSeek-V4-Flash-Vision-Exp, revision 86f746b36186f0e567729a5c06a8c918caba82a9, served as deepseek-v4-flash-vision-exp by a patched vLLM 0.25.2.dev0+g752a3a504.d20260714 runtime with FP8 weights, an NVFP4 key-value cache and a 1,048,576-token context limit. Phi-4 8-bit runs used Transformers 4.56.1 and bitsandbytes 0.47.0. We do not equate this checkpoint with any other DeepSeek release. The matched condition retained the historical 700-token output cap; the native and positive-control conditions froze a 4,096-token output cap. Reasoning text was returned separately in message.reasoning; a separate reasoning-token count was recorded only if exposed by the service.

A separately prospectively frozen PHI-BUDGET-SENSITIVITY condition retained the historical Phi-4 revision, 4-bit runtime, prompt semantics, primitive vocabulary, worlds, three candidate slots, 16 plans per slot, J0-J5 gates, interventions and hidden-information boundaries. Its only scientific change was increasing the predeclared completion cap from 700 to 2,048 tokens; it did not add representation attempts, candidates or interventions and is not compute-matched to the historical condition. The source protocol was frozen at commit 4606413 and tag nmi-extension-v1-protocol-freeze. Complete known-family (n=400) and held-out (n=100) shards were incorporated after shard completion and before analysis tables were generated through amendment NMI-MIN-SENS-V1-AMENDMENT-002. The fixed 96-world panel provides the primary paired budget comparison, while the complete populations are reported descriptively.

The superseded broad protocol stopped after eight of 38 planned shards reached complete_verified. Besides the two Phi-budget shards, six completed descriptive shards tested matched DeepSeek, Phi-4 strict-schema decoding and Phi-4 one-repair on 400 known-family and 100 held-out worlds each; all six produced zero successful worlds. Strict-schema Phi-4 yielded 28/19,200 executable known-family plans and 1/4,800 held-out plans, while the other completed shards yielded none. Three incomplete runs were retained but excluded: native DeepSeek known-family (404 call records), Phi-budget control (1,043) and an infrastructure-terminated Phi-budget attempt (202/2,400). Reporting these superseded runs prevents selective omission; they are not treated as a completed factorial or replacement confirmation.

Worlds, primitive vocabulary, three candidate slots, 16 four-step plans per slot, representation opportunities, J0-J5, prospective interventions and hidden-information boundaries were unchanged. The matched condition received no additional semantic information. The positive control fixed only the correct representation; the model still authored the executable expression, explanation and intervention selection, so success could not be created by overwriting those fields. Because the offline historical cascade had 100% pre-executable attrition, exceeding the frozen 25% trigger, one additional Phi-4 8-bit condition permitted exactly one replacement repair response per structurally invalid slot. Validator feedback named only syntax, schema, operation, reference, type or arity errors and exposed no truth, target distance, outcome or gate value.

Grammar-constrained interface sensitivity

After auditing the frozen responses, we specified one additional DeepSeek condition on the identical 96-world panel. Each of three slots retained 16 independent four-step plans and therefore 48 representation opportunities per world. The first call reserved 4,096 tokens for native reasoning; a second call received that model-produced deliberation and a complete parser-level operation syntax, used reasoning_effort=none, and reserved 4,096 tokens for a compact grammar-constrained plan object. These two calls replaced the original proposal-plus-explanation allocation, preserving six calls per world. The schema guaranteed only JSON structure, plan count, depth, operator names and argument keys/types. The grammar-constrained condition exposed 28 non-crossover primitives because no donor graph was supplied; the original CJ5 vocabulary contained 29 including crossover.

The serializer received no fitted expression, truth, target distance, intervention outcome, gate value or semantic repair. Worlds, primitive vocabulary, candidate slots, opportunities, deterministic downstream fitter and intervention selector, hidden-information boundaries and J0-J5 were unchanged. The model supplied graph-edit topology; the scaffold supplied motif realization, fit, selection among the 16 plans and the committed maximum-separation intervention. The protocol and code were frozen at commit b6e1561 and tag nmi-fair-interface-v1-protocol-freeze. A subsequent operational amendment partitioned families into four disjoint shards without changing calls or scientific settings. A post-freeze 31-call throughput pilot used eight formal-panel worlds and returned 15 non-empty responses, but produced no candidate, world or summary table; it was archived and excluded before formal shards began, and the unchanged worlds were rerun. Concurrent shard launch then exposed a four-sequence server limit: three timeout-only shard directories contained no returned response or result table. We archived and excluded them, froze a second operational amendment at commit 6bab5fb and tag nmi-fair-interface-v1-sequential-shards, and scheduled the unchanged shards sequentially.

Worlds, representations and search

Generators returned public observations, candidate intervention actions without outcomes, a frozen incumbent LanguageSpec, hidden truth and independent intervention and falsification cases. AJ5 used eight families and 50 jump seeds per family; CJ5 used 50 new seeds per family. No-jump controls used 25 seeds per family in each phase. CJ5 added 100 held-out jump and 100 control seeds. Representations were canonical typed graphs with executable fitted programs.

AJ5 tested direct model output, model sampling, fixed-space reasoning, attribute-only mutation and two typed high-level proposal conditions. CJ5 used 29 generic local rewrites. C3 traversed 48 four-operation branches and retained three candidates by a deterministic score. The shared realizer matched candidate graphs against nine hand-authored motifs, assigned fixed basis functions aligned with the procedural mechanisms and solved only their coefficients from public observations. The held-out triadic-product basis was present before unlock. C_rand sampled 48 four-step paths. C_self requested 16 plans in each of three slots; invalid plans were not repaired.

Prospective evaluation

J0 required incumbent observation MSE<=10-12; J1 required DSL validity and frozen-language non-membership; J2 required candidate observation MSE<=10-12; J3 required candidate-oracle prediction separation>=0.5 on the committed action; J4 required intervention MSE improvement>0.1; J5 required falsification MSE<=10-12 and improvement>0.1. Candidate, prediction, action and split hashes were committed before simulator evaluation.

Statistics and component audit

The world was the replicate. AJ5 used 10,000 family-stratified paired bootstrap replicates for effect uncertainty; its archived uncentered bootstrap tail proportions are not null-calibrated tests. CJ5 used the same bootstrap for effects and paired sign-flip tests. Holm correction was applied within prospectively specified comparison families at $\alpha=0.05$. Candidate rows were used only for attrition. Family-level distributions are reported descriptively because eight known families plus one held-out family do not support a stable population-level family inference.

The post-hoc component audit reconstructed all C3 candidates from archived deterministic representations, fitted expressions and selected interventions, replacing the model-derived explanation with an empty string. It then regenerated each world, froze a new commitment and recomputed J0-J5. No model inference was run.

The realizer-dependence protocol was frozen at commit 7753db8 and tag nmi-realizer-audit-v1-protocol-freeze before formal counterfactual replay. It fixed 3,288 archived candidate slots: C3 and C_rand on all 500 jump worlds and grammar-constrained DeepSeek on the 96-world panel. Eleven policies comprised aligned replay, complete motif disabling, role/action-blind field rebinding and one mask for each of eight non-incumbent signatures. Complete disabling replaced motif-derived expressions with the exact incumbent expression while preserving the candidate graph. Role/action-blind replay preserved motif algebra but assigned type-compatible public fields in lexical order, refitted coefficients from public observations and did not inspect node roles or intervention-key changes. Signature masks used incumbent fallback only for the named detected motif. Every policy deterministically selected the public action with maximum candidate-incumbent prediction separation before hidden outcomes were evaluated. The audit made zero model calls, treated worlds as the analysis unit and used candidate rows only for gate attrition.

For the targeted sensitivity analysis, exact world counts, JSR and Wilson 95% intervals are primary. Comparisons use paired world transitions and paired JSR differences on the identical 96-world panel; family rates are descriptive. The full n=400 known-family and n=100 held-out Phi-4 budget populations are supplementary descriptive sensitivities, and the n=40 positive control is shown separately. No candidate-level significance tests are performed and P values are not used as primary evidence. Response and plan attrition are reported with their denominators, followed by executable-candidate J1-J5 attrition.

Integrity and replay

Requests, representations, ancestry edges, fitted programs, commitments, gate values and configuration hashes were retained. Replay reconstructed representations and recomputed J0-J5. Three grammar-constrained-interface shard launches were interrupted after queue starvation produced transport timeouts but before any response or scientific result was returned; their logs were archived outside the formal namespace under a frozen operational amendment, and the unchanged shards were restarted sequentially. No completed shard was rerun and there were no outcome-quality exclusions.

AI assistance in research and writing

Phi-4 and DeepSeek generated the experimental outputs described above. OpenAI ChatGPT and Codex were used from 2 to 5 September 2026 to discuss methodological alternatives, stress-test interpretations, inspect artifacts, implement and orchestrate separately frozen analyses, verify literature metadata and assist manuscript drafting; the conversational service did not expose a stable deployed-build identifier, so no finer model-version claim is made. These systems did not alter historical confirmatory source data or change frozen evaluation rules. All protocols, code changes, statistical outputs, citations and interpretations were reviewed and approved by the human authors, who retain full responsibility for the work.

Data availability

Synthetic-world definitions, historical and extension result tables, manifests and replay artifacts, including the inference-free realizer audit, are included in the project repository (https://github.com/gbanyan/abductive-jump). Historical AJ5/CJ5 raw call ledgers and raw superseded-extension shards are not tracked directly in Git; a release utility combines them with the tagged publication commit, writes a file-level SHA-256 manifest and stream-verifies the archive. The exact submission snapshot and its checksum are provided in release nmi-github-submission-v4 (https://github.com/gbanyan/abductive-jump/releases/tag/nmi-github-submission-v4). The original confirmatory state and separate sensitivity namespaces are identified by commits and tags. No personal or restricted third-party data were used. A DOI-minted archival copy will be deposited before publication.

Code availability

Source code for generation, search, evaluation, offline attrition, statistics, figure generation, component audit, replay and reviewer-archive verification is included in the public project repository and fixed by release nmi-github-submission-v4. Original software is licensed under Apache-2.0 and original synthetic research data and derived artifacts under CC BY 4.0; path-level scope is stated in LICENSE_SCOPE.md. A DOI-minted archive will be cited in the accepted version. Model weights are not redistributed; repository identifier, revision and runtime are reported above.

References

1. Peirce, C. S. Deduction, induction, and hypothesis. Popular Science Monthly 13, 470-482 (1878).

2. Harman, G. H. The inference to the best explanation. Philos. Rev. 74, 88-95 (1965).

3. Lipton, P. Inference to the Best Explanation (Routledge, 2004).

4. Boden, M. A. The Creative Mind: Myths and Mechanisms (Routledge, 2004).

5. Wiggins, G. A. A preliminary framework for description, analysis and comparison of creative systems. Knowl.-Based Syst. 19, 449-458 (2006).

6. Wiggins, G. A. Searching for computational creativity. New Gener. Comput. 24, 209-222 (2006).

7. Muggleton, S. & Buntine, W. Machine invention of first-order predicates by inverting resolution. In Proc. 5th Int. Conf. Machine Learning 339-352 (1988).

8. Stahl, I. The appropriateness of predicate invention as a bias shift operation in inductive logic programming. Mach. Learn. 20, 95-117 (1995).

9. Donoho, S. K. & Rendell, L. A. Rerepresenting and restructuring domain theories. J. Artif. Intell. Res. 2, 411-446 (1995).

10. Schmidt, M. & Lipson, H. Distilling free-form natural laws from experimental data. Science 324, 81-85 (2009).

11. Udrescu, S.-M. & Tegmark, M. AI Feynman. Sci. Adv. 6, eaay2631 (2020).

12. Fawzi, A. et al. Discovering faster matrix multiplication algorithms with reinforcement learning. Nature 610, 47-53 (2022).

13. Romera-Paredes, B. et al. Mathematical discoveries from program search with large language models. Nature 625, 468-475 (2024).

14. Wang, R. et al. Hypothesis Search: inductive reasoning with language models. In ICLR (2024).

15. Zhou, Y. et al. Hypothesis generation with large language models. In Proc. 1st Workshop on NLP for Science 117-139 (Association for Computational Linguistics, 2024).

16. Huang, K. et al. Automated hypothesis validation with agentic sequential falsifications. In Proc. 42nd International Conference on Machine Learning, PMLR 267, 25372-25437 (2025).

17. Lu, C. et al. The AI Scientist. Preprint at https://arxiv.org/abs/2408.06292 (2024).

18. Gottweis, J. et al. Accelerating scientific discovery with Co-Scientist. Nature 655, 487-496 (2026). https://doi.org/10.1038/s41586-026-10644-y

19. Ghareeb, A. E. et al. A multi-agent system for automating scientific discovery. Nature 655, 497-505 (2026). https://doi.org/10.1038/s41586-026-10652-y

20. Boiko, D. A. et al. Autonomous chemical research with large language models. Nature 624, 570-578 (2023).

21. Bellemare-Pepin, A. et al. Divergent creativity in humans and large language models. Sci. Rep. 16, 1279 (2026). https://doi.org/10.1038/s41598-025-25157-3

22. Zahavy, T. LLMs can't jump. PhilSci-Archive preprint 28024 (2026). https://philsci-archive.pitt.edu/28024/

23. Pu, Y., Lin, T. & Chen, H. Principle-Evolvable Scientific Discovery via Uncertainty Minimization. In ICML, PMLR 306 (2026).

24. Murphy, K. Model Discovery Agent: LLM-assisted Bayesian experiment design for data-efficient discovery of mechanistic world models. Preprint at https://arxiv.org/abs/2608.09696 (2026).

25. Zhong, T. et al. Before the Action: Benchmarking LLMs on Prospective Hypothesis Discovery. Preprint at https://arxiv.org/abs/2607.15766 (2026).

26. Zhao, Q. et al. EvoSCM: scientific belief revision through causal model evolution and experimentation. Preprint at https://arxiv.org/abs/2609.01526 (2026).

27. Chen, T. et al. HypoSpace: a diagnostic benchmark for set-valued hypothesis generation under underdetermination and sublinear coverage bounds. Preprint at https://arxiv.org/abs/2510.15614 (2025).

28. Liu, Y. et al. ResearchBench: benchmarking LLMs in scientific discovery via inspiration-based task decomposition. In Findings of the Association for Computational Linguistics: ACL 2026 13187-13207 (2026). https://doi.org/10.18653/v1/2026.findings-acl.644

29. Ding, A. W. & Li, S. Generative AI lacks the human creativity to achieve scientific discovery from scratch. Sci. Rep. 15, 9587 (2025). https://doi.org/10.1038/s41598-025-93794-9

30. Pearl, J. Causality 2nd edn (Cambridge Univ. Press, 2009).

31. Peters, J., Janzing, D. & Schölkopf, B. Elements of Causal Inference (MIT Press, 2017).

Acknowledgements

No additional acknowledgements.

Author contributions

J.-R.H. performed conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, project administration and writing of the original draft. W.-H.L. performed supervision and writing - review and editing. Both authors reviewed and approved the manuscript.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests

The authors declare no competing interests.

Correspondence

Correspondence and requests for materials should be addressed to Jing-Rung Huang (p78084063@mail.ncku.edu.tw).

Supplementary Information

Contents: Supplementary Methods S1-S18 (including Supplementary Table 1); targeted sensitivity source tables; Extended Data Figures 1-3 (world success, attrition, per-family results); realizer-audit source tables. Conditional rates with no executable candidates are NA.

Supplementary Methods

S1. Formal estimand

Let R0 denote the incumbent representation and A0 the set of representations admitted by the frozen LanguageSpec. Let P0 = world.incumbent_programs denote the finite frozen executable comparator set. The oracle selects h0* = argmin_h (L_obs(h), canonical_json(h)) over P0, using lexicographic ordering of the pair. Thus ties in observation loss are broken deterministically by canonical JSON, not by intervention performance. We do not assume that all observational optima are intervention-equivalent, nor claim a universal optimum over every function expressible by the language. J4 and J5 compare against this selected observational optimum.

A candidate theory contains a representation R and an executable expression f. J1 requires a valid representation with R outside A0; the later gates evaluate f. The canonical membership certificate concerns explicit structural constraints, not a general proof that no functionally equivalent incumbent expression exists. A world-level validated jump requires at least one of three candidates to pass all J0-J5 gates.

The jump success rate is the proportion of jump worlds with a validated candidate. The false-jump rate applies the identical decision to control worlds whose truth is in the frozen incumbent program set. Abductive precision is the number of validated candidates divided by candidates passing J0-J3. These quantities answer different questions and are never substituted for one another.

S2. Procedural families

AJ5 generated eight families: latent common cause, unification, hidden regimes, property-to-relation, state invention, coordinate transformation, causal ambiguity and meta-law. Each generator returned public observations, an incumbent representation and language, a hidden executable truth, a finite intervention domain and separate held-out falsification cases. Generator validation checked deterministic output, truth redaction, exact incumbent observational fit, existence of a discriminating intervention and reachability of a valid escaped explanation.

Jump seeds were 10000-10049 per family and controls 20000-20024 per family. CJ5 used reconstruction seeds 30000-30049, controls 50000-50024, held-out jump seeds 40000-40099 and held-out controls 60000-60099. World generation used no LLM output.

S3. Typed representation and incumbent membership

A representation is a canonical graph of typed nodes and edges plus executable equations, dependencies, constraints, observability, arity, temporal indices and argument bindings. Canonical serialization sorts unordered fields before hashing. LanguageSpec.contains checks node kinds, edge/relation forms, equation families and other frozen admissibility constraints. J1 records at least one concrete membership failure and cannot be satisfied by renaming or paraphrase. The public world supplied candidate intervention actions and input settings to proposers and fitters, but withheld every intervention outcome.

The implementation exhausts world.incumbent_programs, sorts by observation loss and canonical program JSON, and returns the first program. Exactness refers to this finite comparator set. Any failure of J0 invalidates a world rather than removing it as difficult. No such confirmatory failure occurred.

S4. AJ5 conditions

- B0 direct LLM: directly emits a typed theory graph.
- B1 sample matched: high-temperature independent samples propose at most three typed operations.
- B2 fixed-space agent: representation remains R0; calls may reason and revise inside it.
- B3 attribute mutation: an external mechanism changes only values/equation attributes allowed by the frozen incumbent language.
- B4 representation mutation: three proposals sampled with replacement from a nine-member typed portfolio.
- B5 full system: three structurally distinct portfolio proposals, archive accounting and deterministic falsification.

The portfolio contained latent, invariant, sign-contrast regime, additive relation, additive state, square-function, affine-context, causal-edge and transition variants. This high-level alignment motivated CJ5 and is not hidden as a limitation.

The P0/P1/P2 factorial reused the same reasoning code and world populations. Only the proposal source differed: LLM, external portfolio or oracle-correct representation. At the proposal stage, P2 received the target representation but not fitted coefficients, truth-program output or hidden outcomes. Downstream fitting and maximum-separation intervention selection were supplied by the deterministic scaffold, shown in the second call and enforced in the evaluated artifact. The later DeepSeek supplied-representation sensitivity is a distinct control: its expression and selected intervention are model-authored.

B2 and B3 are structural negative controls that cannot pass J1 by design; their zero JSR does not establish a general search or reasoning limitation.

S5. CJ5 generic primitive language

GENERIC_PRIMITIVE_SET_V1 contains 29 local operations: add/remove node, edge, function, equation, dependency, constraint and temporal index; reverse/retype/relabel; change observability or arity; bind/unbind; compose/decompose; merge/split; reify edge or node; and bounded copy/crossover. ADD_NODE creates an untyped Primitive; typing, observability, arity, temporal indexing, dependencies and each argument binding require separate records.

Every record binds the parent hash, child hash, operator, canonical arguments, seed and depth. The implementation rejects family names, truth, outcome fields, target distance and family-level aliases such as LATENTIZE, ADD_STATE, PROPERTY_TO_RELATION, ADD_REGIME, COMMON_CAUSE, META_LAW and COORDINATE_TRANSFORM.

S6. CJ5 search and controls

C3 traversed 48 branches of four local edits. Fixed allocations covered generic topologies for typed nodes, function composition, argument-bound functions, relations and reified edges. Ranking used only DSL validity, incumbent non-membership, observational compatibility, existence of an outcome-blind discriminating query, structural novelty and ancestry diversity.

C0 evaluated 192 within-space candidates. C2 evaluated 192 depth-one alternatives. C_rand drew 48 four-step paths and selected final candidates by seed-fixed structural hash. C_self supplied the identical manifest to Phi-4 and requested 16 plans in each of three slots; invalid plans consumed capacity and were not externally completed. C1 retained the AJ5 atomic portfolio as a reference. C5 supplied the target representation and deterministic truth compiler as a conditional ceiling. Because the operation semantics of C1 and C5 differ, they are shown separately in cost comparisons.

C3 and C_rand differ in both traversal and final selection. Their contrast estimates a difference between complete search-and-selection policies, not a proposal-only effect.

S7. Held-out family and lock

triadic_relation_reification uses three correlated observed inputs. The incumbent language fits a cubic rule on the observational support. The target represents an arity-three relation whose product mechanism separates under intervention. A valid construction requires edge reification, an arity change and separate argument bindings. No primitive produces the complete target.

The generator, structure and confirmatory seeds were excluded from AJ5 and CJ5 pilot inference. Only deterministic unit tests for redaction, oracle adequacy and constructive reachability preceded the lock. Known-family and control runs were terminal before the held-out unlock commit. Because AJ5 contained a binary property-to-relation family and the generic set already included reification, this is structural-family hold-out, not wholly concept-free invention.

S8. Candidate fitting and intervention selection

In structured_search, every intermediate graph is realized and fitted by evaluate_candidate before its score is computed. The final _select_diverse ranks terminal evaluations using those fitted scores and structural diversity, retaining three. There is no fit-after-retention stage supplying the initial ranking scores. The selected candidates are then committed before outcome reveal.

The shared compositional realizer does not fit an unrestricted model to the candidate graph. It matches the graph against nine pre-specified structural signatures, assigns the fixed basis functions below and solves their coefficients from public observations. It cannot read hidden truth, validation outcomes or family labels, but the basis library was hand-authored to cover the procedural mechanisms. The held-out triadic-product basis existed at the CJ5 freeze before held-out unlock.

Structural signature	Fixed basis supplied by the realizer	Aligned procedural mechanism
relation_arity_3	x·z·w	held-out triadic product
temporally_indexed_recurrence	x, history sums	state invention
unobserved_dependency	raw stable proxy	latent common cause / causal ambiguity
unobserved_selector	regime-signed x	hidden regimes
bound_relation	observed scalar variables	property-to-relation
self_composed_function	x^2	coordinate transformation
shared_rule_binding	x, context	unification
multi_argument_function	x, x·context	meta-law
incumbent_basis	first two scalar variables	fallback incumbent form

For J3, the deterministic evaluator compares realized candidate and incumbent predictions over the finite public action set and selects their maximum absolute separation. It records action, predictions, candidate hash and split hash before querying the simulator. Thus C3 and the grammar-constrained interface condition supply graph topology to a scaffold that supplies the basis, fit, ranking and committed intervention. Model-proposed actions are retained only as diagnostics in the C-series conditions; the separate DeepSeek P2 control instead evaluates its model-authored expression and selected intervention.

Thresholds were frozen: J0 and J2 observation MSE<=10-12; J3 separation>=0.5; J4 candidate intervention MSE < incumbent MSE-0.1; J5 candidate falsification MSE<=10-12 and < incumbent MSE-0.1. All six gates are conjunctive.

S9. Opportunity and compute accounting

Every condition-world cell had three final slots, two LLM calls per slot, one final candidate evaluation and one intervention commitment per slot, with 700 completion tokens per call. AJ5 therefore executed 32,400 prospectively specified calls plus 3,600 A6 calls. CJ5 executed 33,600 calls. Unused completion capacity was not reassigned. Actual calls, tokens, attempted/valid operations, candidate evaluations, interventions and latency are reported rather than claiming equality of realized token counts.

S10. Statistical analysis

The offline comparator check (scripts/audit_nmi_control_comparator.py) regenerated archived control worlds using their recorded family and seed. The selected observation-optimal program matched the simulator exactly on all intervention and falsification cases: 1,125 cases in 200 AJ5 controls, 1,125 in 200 known-family CJ5 controls, and 500 in 100 held-out CJ5 controls. This verified finite-domain condition, not merely truth membership or observation optimality, prevents strictly positive error reduction in these controls. It does not assert intervention-equivalence of observational optima in general.

The replicate was the world. AJ5 used 10,000 deterministic, family-stratified paired bootstrap replicates with seed 20260902. Seeds were resampled inside each family and conditions remained paired; the aggregate gave each family equal weight. Percentile 95% intervals were reported for rates and paired differences. For each pair, the statistic was the equal-family mean of within-world binary success differences. Replicates resampled paired seeds with replacement separately within each family. The RNG seed was 20260902 XOR the sum of character codes in the concatenated left/right condition labels. The archived field p_one_sided equals (1 + number of bootstrap differences at or below zero) / 10001; p_holm applies the step-down Holm arithmetic to eight such quantities. These uncentered bootstrap samples estimate effect uncertainty, not a distribution constructed under the null of zero difference. We therefore describe those unchanged historical fields as bootstrap tail proportions, not calibrated hypothesis-test P values; no replacement test or model run was introduced.

CJ5 used the same stratified bootstrap for effect intervals and paired random sign-flip tests for prospectively specified beneficial contrasts. C3-C0 and C3-C2 formed the known-family primary multiplicity family; C3-C1, C3-C_rand and C3-C_self formed a secondary family. Held-out contrasts formed their own prospectively specified family. Two-sided Wilson intervals summarize JSR/FJR. No candidate-level pseudoreplication was used.

Post-hoc component audit

An inference-free audit replaced both C3 model calls with a valid empty explanation while preserving only the archived deterministic representation, fitted expression and maximum-separation intervention. Every world was regenerated and every commitment and J0-J5 verdict recomputed. All 2,400 candidate verdicts matched, retaining 500/500 jump successes and 0/300 control false jumps. The audit is explicitly post-hoc and changes attribution, not the commit-frozen comparison.

In the 400 known-family worlds, all 1,200 historical C_self responses were non-empty and the frozen legacy parser extracted a JSON object, but the outer plans value was never a list. Thus 0/19,200 prospectively specified plan opportunities reached execution or structural evaluation. Every response reached the 700-token cap and strict whole-response JSON validity was 0/1,200. Incumbent fallback candidates inserted after empty self-search are excluded from proposal attrition. The analysis made no model calls.

Retained jump gain was rho_J=(JSR_C3-JSR_C0)/(JSR_C1-JSR_C0). It is undefined for a non-positive denominator. It compares observed rates and does not normalize unequal operator semantics.

S11. Exclusions, failures and replay

Parse errors, invalid graphs, non-exact fits, absent discriminating interventions and failed falsification were outcomes, not exclusions. A whole shard could be rerun only for a configuration/hash mismatch, unavailable server, missing/duplicate rows or infrastructure interruption, using identical seeds and model configuration. Confirmatory exclusions and shard reruns were both zero.

Replay rebuilt each selected representation from R0 and its ancestry, recomputed canonical hashes and fitted expressions, reconstructed the frozen intervention, and recomputed J0-J5. AJ5 reproduced 10,800/10,800 selected candidates; CJ5 reproduced 16,800/16,800 with 35,533 ancestry records. There were no mismatches.

Minimal targeted sensitivity protocol

Historical Phi-4 results were preserved on a dedicated tag and archive branch. The new protocol used a separate namespace and a commit-frozen panel selected by outcome-blind salted SHA-256 ranking. The same 12 existing seeds were applied to each of eight known families, yielding 96 paired worlds; no new generator or seed was introduced. A balanced supplied-representation positive control used five of those seeds per family (n=40).

The only new model conditions were Phi-4 8-bit C_self, DeepSeek matched C_self (reasoning_effort=none, 700 output tokens), DeepSeek native C_self (reasoning_effort=max, 4,096 output tokens) and DeepSeek supplied-representation positive control (reasoning_effort=max, 4,096 output tokens). Offline attrition crossed the prospectively frozen 25% pre-execution trigger, enabling one Phi-4 8-bit repair condition. The repair replaced, rather than augmented, a structurally invalid response and received only validator error classes; it could not receive semantic or simulator feedback.

The separately frozen PHI-BUDGET-SENSITIVITY reused the exact historical Phi-4 revision and 4-bit runtime. Prompt semantics, primitive vocabulary, known and held-out worlds, three candidate slots, 16 four-step plans per slot, representation opportunities, interventions, J0-J5 and all hidden-information boundaries were unchanged. Only generation.max_tokens increased from 700 to 2,048. The condition therefore tests sensitivity to the original completion cap; it is not compute-matched. Its source protocol was frozen at commit 4606413 and tag nmi-extension-v1-protocol-freeze. The completed known-family (n=400) and held-out (n=100) shards were incorporated after shard completion and before analysis tables were generated through amendment NMI-MIN-SENS-V1-AMENDMENT-002; the preselected 96-world panel is the primary paired comparison and the full shards are descriptive.

The original 38-shard broad-extension matrix was superseded after eight shards reached complete_verified. The two Phi-budget shards above were incorporated descriptively. Six additional completed shards were preserved but not used as a completed factorial:

Superseded condition	Population	World successes	Executable plan opportunities
matched DeepSeek	known-family n=400	0/400	0/19,200
matched DeepSeek	held-out n=100	0/100	0/4,800
Phi-4 strict-schema decoding	known-family n=400	0/400	28/19,200
Phi-4 strict-schema decoding	held-out n=100	0/100	1/4,800
Phi-4 one-repair	known-family n=400	0/400	0/38,400
Phi-4 one-repair	held-out n=100	0/100	0/9,600

Three partial runs were also preserved and excluded from inference: native DeepSeek known-family with 404 completed call records, Phi-budget known-control with 1,043 records and an infrastructure-terminated Phi-budget known-family attempt with 202 of 2,400 calls. These records are disclosed for completeness and must accompany the DOI archive; no partial run contributes a denominator or scientific estimate.

Each shard had to match its frozen seed panel, model identity, revision, call cardinality and artifact schema before receiving complete_verified status. Replay was locked until all five shards were verified. Analysis was then locked until all outputs replayed with zero mismatches. Reported inference consists of counts, Wilson intervals, paired world transitions, paired JSR differences and per-family descriptions. DeepSeek native is explicitly not compute-matched, and the Phi-4 sensitivity changes serving engine together with precision.

Conditional attrition rates with denominator zero are reported as NA (no executable candidates); positive rates below 0.1% retain three decimal places. Archived raw tables are preserved; publication tables apply this display convention.

S12. Minimal sensitivity results and attrition

All legacy-interface autonomous fixed-panel conditions produced 0/96 successful worlds (Wilson 95% interval 0-3.8%): the historical Phi-4 4-bit slice, Phi-4 4-bit with the 2,048-token cap, Phi-4 8-bit, DeepSeek matched, DeepSeek native and Phi-4 8-bit with one validator-only repair. The five prospectively specified paired contrasts therefore contained 96 joint failures, no successes unique to either member and a paired JSR difference of 0.000. Each of the eight families contributed 12 worlds and had 0/12 in every legacy-interface autonomous condition; these family counts are descriptive.

The fixed-panel 2,048-token Phi-4 run produced 993/4,608 schema-valid plan opportunities but no executable plan. Across the complete descriptive known-family population, 4,642/19,200 opportunities were schema-valid and 9 executable. Those 9 plans occurred within a single selected slot, yielding one candidate, which failed J1. The held-out population produced 1,289/4,800 schema-valid and 0 executable opportunities. JSR was 0/400 known-family and 0/100 held-out, matching the historical world-level counts.

Phi-4 8-bit produced 4,608/4,608 JSON-extractable effective opportunities but 0 schema-valid because the extracted outer plans field was not a list. Its one-repair condition repeated the same error in every effective replacement opportunity. DeepSeek matched had 4,480 non-list plans errors and 128 missing plans among 4,608 opportunities. DeepSeek native exposed reasoning text in all 576 calls, but no separate answer reached the parser; 518 calls ended at the 4,096-token cap and 58 reported a stop finish reason.

The balanced DeepSeek supplied-representation control succeeded in 3/40 worlds (7.5%; Wilson 95% interval 2.6-19.9%): 1/5 coordinate-transformation and 2/5 hidden-regime worlds, with 0/5 in each other family. Of 120 model-authored expression-and-intervention candidates, 4 were parse-valid, executable and passed J1-J2; 3 also passed J3-J5. The other calls supplied reasoning text without a usable final answer; 116/120 reached the cap. This is a minimum positive control for occasional supplied-representation use, not a reliable ceiling.

The extension used 576 calls in each unrepaired n=96 C_self condition, 864 in the triggered repair and 120 in P2. Total completion tokens were 231,773 for Phi-4 8-bit, 237,792 for DeepSeek matched, 2,249,518 for DeepSeek native, 433,373 for repaired Phi-4, 485,312 for P2 and 528,976 for the fixed-panel Phi-budget condition. The service exposed reasoning text but not a separate reasoning-token count for DeepSeek. All five new shards and both Phi-budget populations replayed without mismatch: 2,772 candidate rows, zero mismatches and zero replay-time model calls.

S13. Grammar-constrained interface sensitivity protocol

An offline audit of the matched and native DeepSeek proposal calls made no model calls. It found that the frozen self-composition prompt listed the available operators but not the exact argument names expected by the executor. All 288 matched proposal responses used at least one of the incompatible keys source, target or type; none was strict whole-response JSON. The legacy parser could extract an inner JSON object from an incomplete outer response, so object extraction was not evidence of a valid top-level plans object. Native proposal calls contained reasoning but no answer content. These findings motivated a single grammar-constrained sensitivity rather than a broader model factorial.

The grammar-constrained condition reused the fixed 96-world panel, three slots, 16 four-step plans per slot, 28 available non-crossover primitives, deterministic motif realizer and intervention selector, and unchanged J0-J5. The twenty-ninth CJ5 primitive, crossover, was unavailable because no donor graph was supplied. Each slot used two calls to the same served DeepSeek checkpoint. A native-reasoning call had a 4,096-token budget. A second reasoning_effort=none call received only the first call's deliberation, public world and exact parser-level syntax and had a separate 4,096-token answer budget. Its strict JSON schema fixed the outer object, 16-plan count, four-step depth, operator vocabulary and operator-specific argument keys and string types. It did not guarantee that referenced nodes or edges existed when used, that operations composed successfully, or that a resulting representation fit observations or separated intervention predictions.

No truth, target distance, fitted expression, intervention outcome, J3-J5 value or semantic validator feedback entered either call. The two-stage allocation preserved six calls per world by replacing the original proposal-plus-explanation calls; no additional representation opportunity was added. Protocol, configuration and code hashes were frozen at commit b6e1561 and annotated tag nmi-fair-interface-v1-protocol-freeze. An operational amendment at commit a65974f and tag nmi-fair-interface-v1-shard-freeze partitioned the eight families into four disjoint two-family shards while retaining concurrency four inside each runner. The server admitted at most four simultaneous sequences. Concurrent launch starved three shards behind that limit: each accumulated four 600-s timeout records but no returned response, candidate table, world table or summary. Those timeout-only directories were archived outside the formal namespace and excluded; a second operational amendment at commit 6bab5fb and tag nmi-fair-interface-v1-sequential-shards froze sequential execution of the unchanged shards. This changed wall time and scheduling only. A separate post-freeze, pre-amendment 31-call throughput pilot used eight formal-panel worlds and returned 15 non-empty outputs but produced no candidate, world or summary table; it was excluded and the worlds were rerun unchanged.

S14. Grammar-constrained interface sensitivity results

The grammar-constrained interface produced strict whole-response JSON, schema-valid operation names and parser-level argument types in 4,608/4,608 plan opportunities. Dynamic execution succeeded for 3,939/4,608 plans (85.5%), and 280/288 selected candidate slots contained at least one executable proposal. Cumulative selected-candidate attrition was 236/280 at J1, 236/280 at J2 and 21/280 at each of J3, J4 and J5. The 21 validated candidates yielded 15/96 successful worlds (15.6%; Wilson 95% interval 9.7-24.2%). Compared separately with the historical Phi-4 slice, matched DeepSeek and native DeepSeek, each paired table contained 81 joint failures, 15 successes unique to the grammar-constrained condition, no successes unique to the reference and a paired JSR difference of +0.156.

Success was family-concentrated: meta-law succeeded in 9/12 worlds and unification in 6/12; causal ambiguity, coordinate transformation, hidden regimes, latent common cause, property-to-relation and state invention each succeeded in 0/12. All 21 validated candidates had structural signature multi_argument_function; the other selected executable candidates comprised one bound_relation and 258 incumbent_basis realizations. On the same panel, C_rand succeeded in 16/96 worlds: 66 worlds failed under both conditions, 15 succeeded only under C_rand, 14 only under the grammar-constrained condition and one under both. Thus the model-generated plans did not exceed random composition in aggregate and showed a different family-specific motif distribution. Family counts are descriptive. All 288 deliberation calls returned reasoning text and reached the 4,096-token cap. All 288 serialization calls returned strict schema-valid answers and stopped before their separate cap. Deliberation and serialization used 1,179,648 and 574,538 completion tokens, respectively; the service exposed no separate reasoning-token count. The formal condition used 576 calls and no transport retry.

Deterministic replay verified all 288 candidate rows with zero scientific, prompt or request mismatches and zero replay-time model calls. The initially frozen verifier nevertheless reported 576 metadata mismatches because it expected an engine field on each call-ledger row, whereas the frozen runner stores engine metadata in each shard summary's stage manifests. A post-hoc verifier correction preserved the initial report and frozen verifier hash, verified model identity, revision, quantization, engine, engine version and generation settings across all four shard manifests, and omitted only the nonexistent per-call engine comparison. No inference artifact or scientific result changed.

S15. Software and environment

The repository records package versions, configuration hashes, prompt identifiers, model revision, vLLM image digest, GPU class and random seeds in reproducibility manifests. DeepSeek used FP8 weights with an NVFP4 key-value cache under patched vLLM 0.25.2.dev0+g752a3a504.d20260714; Phi-4 8-bit used Transformers 4.56.1 and bitsandbytes 0.47.0. Tests cover world determinism, public-field redaction, DSL membership, primitive legality, budget invariants, reachability, statistics and replay. Historical raw call ledgers and raw superseded-extension shards are included in the public release preservation archive with file-level hashes; they are not tracked directly in Git. The publication phase did not call the model or generate new experimental rows.

S16. Grammar-constrained prompt and analysis templates

The deliberation system instruction was:

Reason about executable representation changes using only the supplied public world and primitive syntax. Do not use or invent intervention outcomes, hidden truth, target distance or gate feedback. A separate call will serialize your reasoning, so prioritize sixteen independent four-step plans and exact references.

The serialization system instruction was:

Return only the compact JSON plan object required by the response schema. Do not explain, repair semantically, use hidden information or add any outcome claim.

The serialization request contained the public world, the first call's deliberation and a syntax manifest listing the 28 available operators, their exact parser-level argument keys, four required steps per plan and 16 required plans. The strict response schema allowed one object with a single plans array; each plan contained exactly four operation objects, and each operation was one of the frozen operator-specific schemas with no additional properties. Full machine-readable schemas and prompt construction are in src/abductive_jump/fair_interface.py; their hashes are in the fair-interface protocol.

World-level Wilson intervals use the two-sided score interval. AJ5 effect intervals resample seeds within each family, keep conditions paired, aggregate families with equal weight and take the 2.5th and 97.5th percentiles of 10,000 deterministic replicates. CJ5 sign-flip tests flip paired world differences under the null, use the finite-replicate correction (extreme+1)/(replicates+1) and apply Holm correction within each prospectively specified comparison family. Candidate rows enter attrition tables only and never replace the world as the inferential unit.

S17. Realizer-dependence counterfactual audit

The post-confirmatory protocol was frozen at commit 7753db8 and annotated tag nmi-realizer-audit-v1-protocol-freeze before formal replay. SHA-256 hashes fixed the CJ5 candidate table, grammar-constrained configuration, call ledger and candidate table, as well as the runner and counterfactual-realizer code. The audit made zero model calls and refused execution if any frozen input or code hash changed.

The audit fixed all three selected candidate slots from C3 and C_rand for 400 known-family and 100 held-out jump worlds, plus all three selected grammar-constrained DeepSeek slots for 96 known-family worlds. It therefore evaluated 3,288 archived candidate slots under eleven policies, producing 36,168 candidate-policy rows and 12,056 world-policy rows. Aligned replay reproduced every archived candidate gate and world verdict with zero mismatches.

motif_disabled replaced the expression of every detected non-incumbent motif with the exact incumbent expression while retaining the archived candidate representation. The maximum-separation action was then reselected from the unchanged public action set and committed before hidden outcomes were evaluated. Cumulative candidate attrition remained 1,168/1,500 through J2 for C3, 1,040/1,500 for C_rand and 236/288 for grammar-constrained DeepSeek, but all three reached 0 at J3. World success consequently changed from 400/400 known plus 100/100 held out to zero for C3, from 52/400 plus 13/100 to zero for C_rand, and from 15/96 to zero for DeepSeek.

Because incumbent substitution sets candidate and comparator predictions equal, maximum separation is zero, below the J3 threshold of 0.5. J3 failure follows by construction. This is a structural negative control for removal of non-incumbent predictive content, not evidence that this specific basis library is irreplaceable. Signature masking has the same built-in effect on affected candidates; world losses describe coverage among the fixed archived slots.

role_action_blind_binding retained the algebraic terms supplied by each detected motif but replaced their variable identities with type-compatible, non-nuisance public fields in lexical order. It did not use representation roles or inspect which fields changed in intervention queries. Coefficients were refitted using public observations. This policy retained 347/400 known-family and 100/100 held-out C3 worlds, 57/400 and 13/100 C_rand worlds, and 8/96 DeepSeek worlds. C3 retained 26/50 hidden-regime and 21/50 meta-law worlds and all worlds in the other seven families; DeepSeek retained 5/12 meta-law and 3/12 unification worlds. Because motif algebra remained intact, this condition isolates field-binding information but is not a semantics-free realizer.

Eight leave-one-signature-out policies replaced only the named motif with incumbent fallback. For C3, masking relation_arity_3 removed all 100 held-out worlds; masking unobserved_dependency removed 100 known-family worlds; and masking each of bound_relation, multi_argument_function, self_composed_function, temporally_indexed_recurrence and unobserved_selector removed 50. Masking shared_rule_binding removed no world because selected multi_argument_function candidates provided redundant coverage in unification worlds. For DeepSeek, only multi_argument_function supported validated candidates, so its removal changed 15/96 to zero. Exact Wilson intervals, paired transitions, per-family counts, signature distributions and gate attrition are stored in experiments/nmi_realizer_audit_v1/analysis.

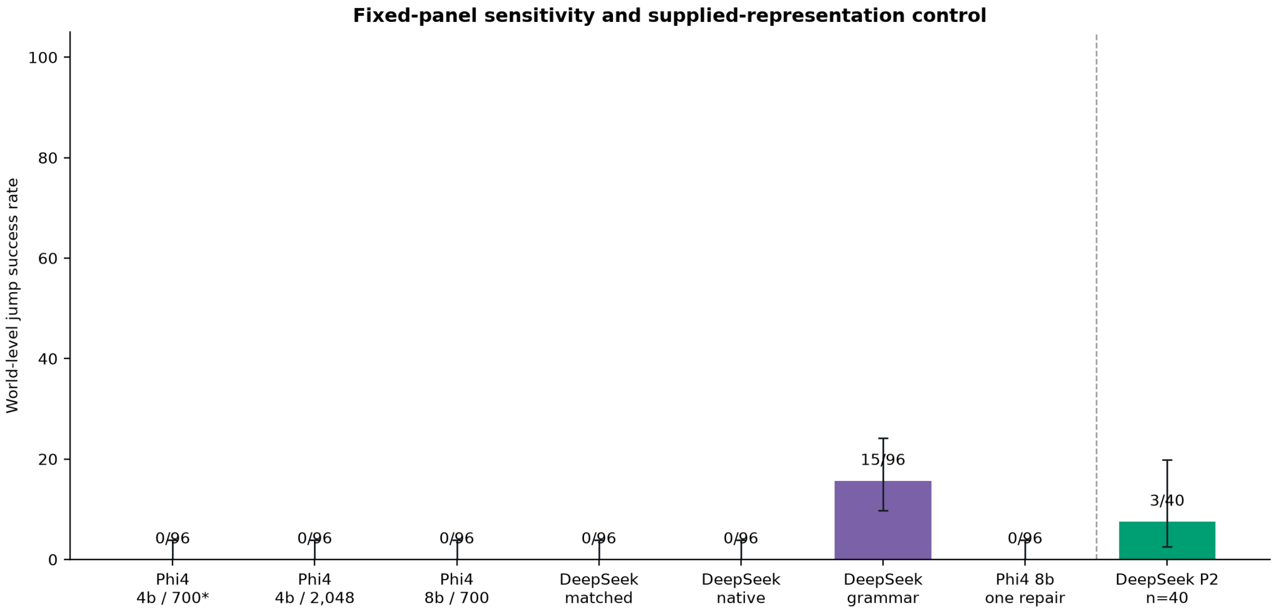
The audit fixes candidates that were originally selected under the aligned realizer and therefore does not estimate how search would adapt if rerun under a different realizer. It is a causal sensitivity of the archived end-to-end verdict, not a replacement confirmatory study, a model-conceptualization test or evidence of external scientific generalization.

S18. Position relative to adjacent discovery evaluations

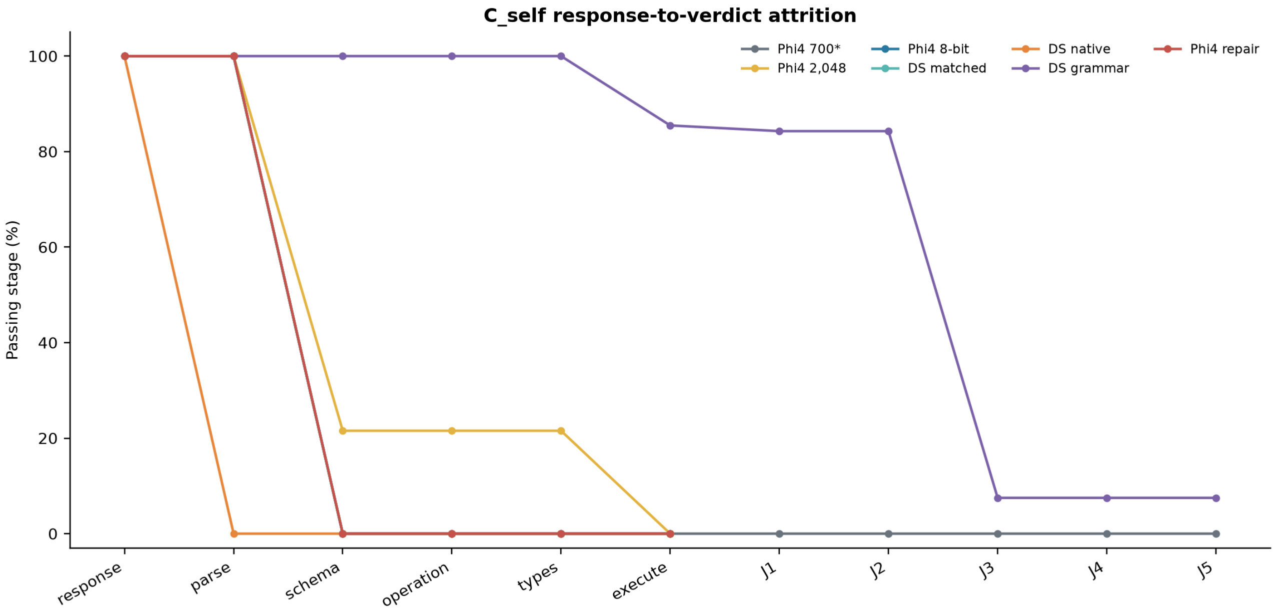
Supplementary Table 1 compares reported evaluation designs. "Not reported" describes the cited publication and does not imply that a feature is absent from unpublished implementations. The comparison is descriptive rather than a priority proof; recent systems already expand principle or model spaces, whereas the present contribution combines a frozen-language certificate, commitment-before-outcome-reveal evaluation, exact replay and component attribution.

Evaluation	Executable hypotheses	Hypothesis-space boundary	Canonical out-of-space test	Prospective test	Held-out check	Component attribution	Replay	Breadth
Hypothesis Search[14]	partial	not reported	not reported	not reported	held-out tasks	hypothesis source / program	not reported	ARC-style induction
HypoGeniC[15]	text hypotheses	not reported	not reported	not reported	held-out examples	generation / ranking	not reported	hypothesis generation
POPPER[16]	partial	not reported	not reported	sequential tests	agentic falsification	hypothesis / validator	not reported	six data domains
FunSearch[13]	yes	fixed program skeleton	not reported	evaluator feedback	held-out tests	proposer / evaluator	partial	mathematics
PiEvo[23]	yes	evolving principle space	not reported	task dependent	task dependent	proposer / search	not reported	four benchmarks, multiple backbones
Model Discovery Agent[24]	yes	open model set	not reported	Bayesian design	posterior checks	proposer / inference	not reported	physics, chemistry and biology
HypoArena[25]	judged text	not reported	not reported	context regression	rubric / judge	not reported	not reported	988 cases, six domains, 15 models
EvoSCM[26]	yes	evolving causal models	not reported	active intervention	prospective prediction	evolution / selection	not reported	simulated physical worlds
This work	yes	frozen formal language	canonical certificate	outcome locked	separate exact cases	proposal-source comparison plus component audit	exact	nine synthetic families; two-checkpoint sensitivity

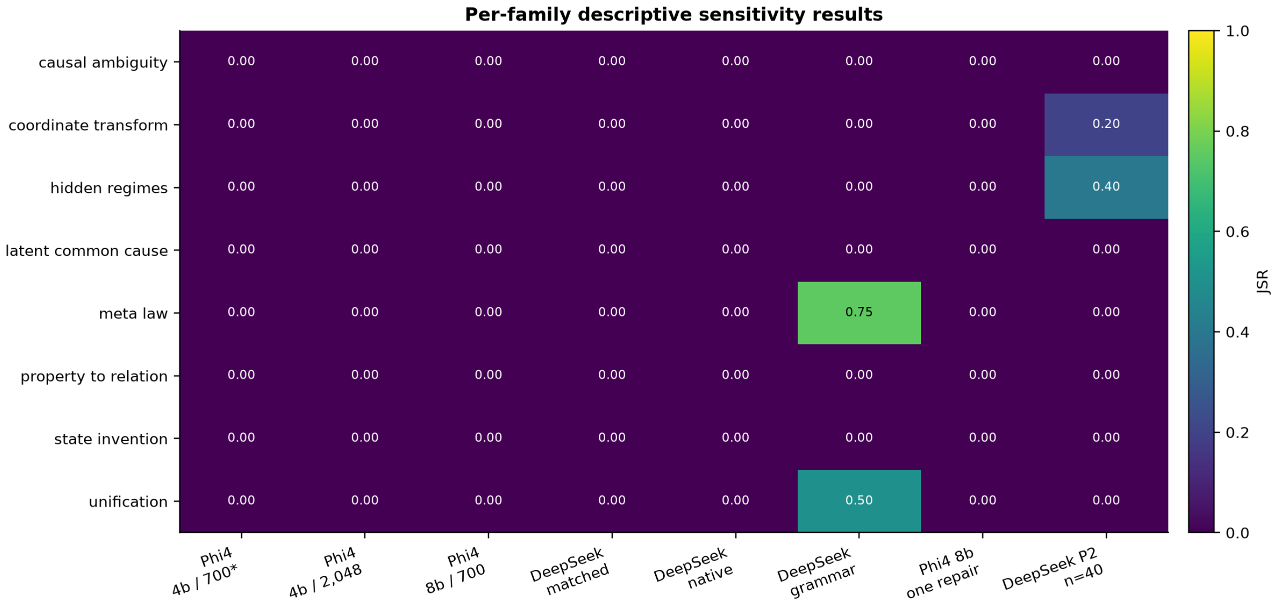
Targeted sensitivity source data



Extended Data Figure 1 | Exact world-level sensitivity results and Wilson 95% intervals, including the separately frozen grammar-constrained interface condition. The asterisk identifies the fixed historical slice of the original n=400 confirmation; the supplied-representation control uses a distinct balanced n=40 subset.



Extended Data Figure 2 | Response-to-verdict attrition, including the grammar-constrained interface cascade. Denominators and units change at the executable boundary and are reported in the accompanying source-data tables.



Extended Data Figure 3 | Per-family descriptive sensitivity results. Grammar-constrained-interface successes were confined to meta-law and unification; the supplied-representation control used a different balanced subset. No family-level or candidate-level significance test was performed.

Sensitivity result table

Condition	Population	Success	JSR	Wilson 95% CI
Historical Phi-4 4-bit C_self	original confirmatory n=400, fixed paired subset shown	0/96	0.0%	0.0-3.8%
Phi-4 4-bit 2,048-token C_self	previously frozen n=400 budget sensitivity, fixed paired subset shown	0/96	0.0%	0.0-3.8%
Phi-4 8-bit C_self	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
DeepSeek matched C_self	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
DeepSeek native C_self	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
Phi-4 8-bit one-repair C_self	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
DeepSeek supplied-representation P2	new balanced n=40 positive-control subset	3/40	7.5%	2.6-19.9%
DeepSeek grammar-constrained C_self	fixed n=96 sensitivity panel	15/96	15.6%	9.7-24.2%

Paired world-level transitions

Reference	Comparison	Both fail	Both pass	New only	Old only	JSR difference
Historical Phi-4 4-bit C_self	Phi-4 4-bit 2,048-token C_self	96	0	0	0	+0.000
Historical Phi-4 4-bit C_self	Phi-4 8-bit C_self	96	0	0	0	+0.000
Historical Phi-4 4-bit C_self	DeepSeek matched C_self	96	0	0	0	+0.000
DeepSeek matched C_self	DeepSeek native C_self	96	0	0	0	+0.000
Phi-4 8-bit C_self	Phi-4 8-bit one-repair C_self	96	0	0	0	+0.000
Historical Phi-4 4-bit C_self	DeepSeek grammar-constrained C_self	81	0	15	0	+0.156
DeepSeek matched C_self	DeepSeek grammar-constrained C_self	81	0	15	0	+0.156
DeepSeek native C_self	DeepSeek grammar-constrained C_self	81	0	15	0	+0.156

Complete cumulative gate attrition

Condition	Unit	Stage	Passed	Denominator	Rate
Historical Phi-4 4-bit C_self	proposal_response	response_returned	1200	1200	100.0%
Historical Phi-4 4-bit C_self	proposal_response	parse_valid	1200	1200	100.0%
Historical Phi-4 4-bit C_self	proposal_response	schema_valid	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	operation_valid	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	argument_type_valid	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	executable	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	J1	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	J2	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	J3	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	J4	0	1200	0.0%
Historical Phi-4 4-bit C_self	proposal_response	J5	0	1200	0.0%
Phi-4 8-bit C_self	effective_plan_opportunity	request_returned	4608	4608	100.0%
Phi-4 8-bit C_self	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
Phi-4 8-bit C_self	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
Phi-4 8-bit C_self	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%
Phi-4 8-bit C_self	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
Phi-4 8-bit C_self	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
Phi-4 8-bit C_self	effective_plan_opportunity	executable	0	4608	0.0%
Phi-4 8-bit C_self	effective_plan_opportunity	representation_constructed	0	4608	0.0%
Phi-4 8-bit C_self	executable_self_proposed_candidate	J1	0	0	NA
Phi-4 8-bit C_self	executable_self_proposed_candidate	J2	0	0	NA
Phi-4 8-bit C_self	executable_self_proposed_candidate	J3	0	0	NA
Phi-4 8-bit C_self	executable_self_proposed_candidate	J4	0	0	NA
Phi-4 8-bit C_self	executable_self_proposed_candidate	J5	0	0	NA
DeepSeek matched C_self	effective_plan_opportunity	request_returned	4608	4608	100.0%
DeepSeek matched C_self	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
DeepSeek matched C_self	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
DeepSeek matched C_self	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%

Condition	Unit	Stage	Passed	Denominator	Rate
DeepSeek matched C_self	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
DeepSeek matched C_self	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
DeepSeek matched C_self	effective_plan_opportunity	executable	0	4608	0.0%
DeepSeek matched C_self	effective_plan_opportunity	representation_constructed	0	4608	0.0%
DeepSeek matched C_self	executable_self_proposed_candidate	J1	0	0	NA
DeepSeek matched C_self	executable_self_proposed_candidate	J2	0	0	NA
DeepSeek matched C_self	executable_self_proposed_candidate	J3	0	0	NA
DeepSeek matched C_self	executable_self_proposed_candidate	J4	0	0	NA
DeepSeek matched C_self	executable_self_proposed_candidate	J5	0	0	NA
DeepSeek native C_self	effective_plan_opportunity	request_returned	4608	4608	100.0%
DeepSeek native C_self	effective_plan_opportunity	non_empty_answer	0	4608	0.0%
DeepSeek native C_self	effective_plan_opportunity	json_parse_valid	0	4608	0.0%
DeepSeek native C_self	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%
DeepSeek native C_self	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
DeepSeek native C_self	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
DeepSeek native C_self	effective_plan_opportunity	executable	0	4608	0.0%
DeepSeek native C_self	effective_plan_opportunity	representation_constructed	0	4608	0.0%
DeepSeek native C_self	executable_self_proposed_candidate	J1	0	0	NA
DeepSeek native C_self	executable_self_proposed_candidate	J2	0	0	NA
DeepSeek native C_self	executable_self_proposed_candidate	J3	0	0	NA
DeepSeek native C_self	executable_self_proposed_candidate	J4	0	0	NA
DeepSeek native C_self	executable_self_proposed_candidate	J5	0	0	NA
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	request_returned	4608	4608	100.0%
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%

Condition	Unit	Stage	Passed	Denominator	Rate
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	executable	0	4608	0.0%
Phi-4 8-bit one-repair C_self	effective_plan_opportunity	representation_constructed	0	4608	0.0%
Phi-4 8-bit one-repair C_self	executable_self_proposed_candidate	J1	0	0	NA
Phi-4 8-bit one-repair C_self	executable_self_proposed_candidate	J2	0	0	NA
Phi-4 8-bit one-repair C_self	executable_self_proposed_candidate	J3	0	0	NA
Phi-4 8-bit one-repair C_self	executable_self_proposed_candidate	J4	0	0	NA
Phi-4 8-bit one-repair C_self	executable_self_proposed_candidate	J5	0	0	NA
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	request_returned	4608	4608	100.0%
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	plan_schema_valid	993	4608	21.5%
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	operation_names_valid	993	4608	21.5%
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	argument_types_valid	993	4608	21.5%
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	executable	0	4608	0.0%
Phi-4 4-bit 2,048-token C_self	effective_plan_opportunity	representation_constructed	0	4608	0.0%
Phi-4 4-bit 2,048-token C_self	executable_self_proposed_candidate	J1	0	0	NA
Phi-4 4-bit 2,048-token C_self	executable_self_proposed_candidate	J2	0	0	NA
Phi-4 4-bit 2,048-token C_self	executable_self_proposed_candidate	J3	0	0	NA
Phi-4 4-bit 2,048-token C_self	executable_self_proposed_candidate	J4	0	0	NA
Phi-4 4-bit 2,048-token C_self	executable_self_proposed_candidate	J5	0	0	NA
DeepSeek supplied-representation P2	model_authored_expression_and_intervention_candidate	parse_valid	4	120	3.3%
DeepSeek supplied-representation P2	model_authored_expression_and_intervention_candidate	executable	4	120	3.3%
DeepSeek supplied-representation P2	model_authored_expression_and_intervention_candidate	J1	4	120	3.3%
DeepSeek supplied-representation P2	model_authored_expression_and_intervention_candidate	J2	4	120	3.3%

Condition	Unit	Stage	Passed	Denominator	Rate
DeepSeek supplied-representation P2	model_authored_expression_and_intervention_candidate	J3	3	120	2.5%
DeepSeek supplied-representation P2	model_authored_expression_and_intervention_candidate	J4	3	120	2.5%
DeepSeek supplied-representation P2	model_authored_expression_and_intervention_candidate	J5	3	120	2.5%

Model-call and compute ledger

Condition	Calls	Attempts	Prompt tokens	Completion tokens	Reasoning text	Latency (s)
Phi-4 8-bit C_self	576	576	704376	231773	0	13971.8
DeepSeek matched C_self	576	576	786804	237792	0	10636.3
DeepSeek native C_self	576	576	839796	2249518	576	156547.1
Phi-4 8-bit one-repair C_self	864	864	1254663	433373	0	26462.2
DeepSeek supplied-representation P2	120	120	183891	485312	120	38836.1
Phi-4 4-bit 2,048-token C_self	576	576	704376	528976	0	40446.1

Reasoning-token counts are reported only when exposed by the serving API; reasoning-text availability is not converted into an inferred token count.

Full Phi-4 completion-budget sensitivity

Condition	Population	Success	JSR	Wilson 95% CI
historical phi4 4bit cself full	original confirmatory n=400	0/400	0.0%	0.0-1.0%
phi4 4bit budget cself full	budget sensitivity n=400	0/400	0.0%	0.0-1.0%
historical phi4 4bit cself heldout	original held-out n=100	0/100	0.0%	0.0-3.7%
phi4 4bit budget cself heldout	budget sensitivity held-out n=100	0/100	0.0%	0.0-3.7%

Full paired completion-budget transitions

Reference	Comparison	Both fail	Both pass	New only	Old only	Difference
historical phi4 4bit cself full	phi4 4bit budget cself full	400	0	0	0	+0.000
historical phi4 4bit cself heldout	phi4 4bit budget cself heldout	100	0	0	0	+0.000

Full Phi-4 completion-budget gate attrition

Population	Unit	Stage	Passed	Denominator	Rate
phi4 4bit budget cself full	effective_plan_opportunity	request_returned	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	non_empty_answer	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	json_parse_valid	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	plan_schema_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	operation_names_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	argument_types_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	executable	9	19200	0.047%
phi4 4bit budget cself full	effective_plan_opportunity	representation_constructed	9	19200	0.047%
phi4 4bit budget cself full	executable_self_proposed_candidate	J1	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J2	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J3	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J4	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J5	0	1	0.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	request_returned	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	non_empty_answer	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	json_parse_valid	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	plan_schema_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	operation_names_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	argument_types_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	executable	0	4800	0.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	representation_constructed	0	4800	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J1	0	0	NA
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J2	0	0	NA
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J3	0	0	NA
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J4	0	0	NA

Population	Unit	Stage	Passed	Denominator	Rate
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J5	0	0	NA

Full Phi-4 completion-budget compute ledger

Population	Calls	Attempts	Prompt tokens	Completion tokens	Latency (s)
phi4 4bit budget cself full	2400	2400	2921915	2216914	169361.1
phi4 4bit budget cself heldout	600	600	696489	546374	41524.5

Grammar-constrained DeepSeek sensitivity

Family	Success	JSR	Wilson 95% CI
causal ambiguity	0/12	0.0%	0.0-24.2%
coordinate transform	0/12	0.0%	0.0-24.2%
hidden regimes	0/12	0.0%	0.0-24.2%
latent common cause	0/12	0.0%	0.0-24.2%
meta law	9/12	75.0%	46.8-91.1%
property to relation	0/12	0.0%	0.0-24.2%
state invention	0/12	0.0%	0.0-24.2%
unification	6/12	50.0%	25.4-74.6%

Paired comparison with archived random composition

Comparison	Both fail	C_rand only	Grammar only	Both pass	Success counts
C_rand vs grammar-constrained DeepSeek	66	15	14	1	16/96 vs 15/96

Selected-candidate realizer signatures

Realizer signature	Executable selected	Validated	Successful worlds
incumbent_basis	258	0	0
bound_relation	1	0	0
multi_argument_function	21	21	15

Grammar-constrained compute ledger

Stage	Calls	Prompt tokens	Completion tokens	Cap hits	Latency (s)
deliberation	288	721860	1179648	288	87421.1
serialization	288	1914226	574538	0	23363.1

Grammar-constrained cumulative attrition

Unit	Stage	Passed	Denominator	Rate
plan opportunity	deliberation_returned	4608	4608	100.0%
plan opportunity	serialization_returned	4608	4608	100.0%
plan opportunity	strict_whole_response_json	4608	4608	100.0%
plan opportunity	json_parse_valid	4608	4608	100.0%
plan opportunity	plan_schema_valid	4608	4608	100.0%
plan opportunity	operation_names_valid	4608	4608	100.0%
plan opportunity	argument_types_valid	4608	4608	100.0%
plan opportunity	executable	3939	4608	85.5%
plan opportunity	representation_constructed	3939	4608	85.5%
executable selected candidate	J1	236	280	84.3%
executable selected candidate	J2	236	280	84.3%
executable selected candidate	J3	21	280	7.5%
executable selected candidate	J4	21	280	7.5%
executable selected candidate	J5	21	280	7.5%

Realizer-dependence source data

Aligned, disabled and role/action-blind policies

Source	Population	Policy	Success	JSR	Wilson 95% CI
C3	heldout	aligned	100/100	100.0%	96.3-100.0%
C3	heldout	motif_disabled	0/100	0.0%	0.0-3.7%
C3	heldout	role_action_blind_binding	100/100	100.0%	96.3-100.0%
C3	known	aligned	400/400	100.0%	99.0-100.0%
C3	known	motif_disabled	0/400	0.0%	0.0-1.0%
C3	known	role_action_blind_binding	347/400	86.8%	83.1-89.7%
C_rand	heldout	aligned	13/100	13.0%	7.8-21.0%
C_rand	heldout	motif_disabled	0/100	0.0%	0.0-3.7%
C_rand	heldout	role_action_blind_binding	13/100	13.0%	7.8-21.0%
C_rand	known	aligned	52/400	13.0%	10.1-16.7%
C_rand	known	motif_disabled	0/400	0.0%	0.0-1.0%
C_rand	known	role_action_blind_binding	57/400	14.2%	11.2-18.0%
DeepSeek_grammar	fixed_known_panel	aligned	15/96	15.6%	9.7-24.2%
DeepSeek_grammar	fixed_known_panel	motif_disabled	0/96	0.0%	0.0-3.8%
DeepSeek_grammar	fixed_known_panel	role_action_blind_binding	8/96	8.3%	4.3-15.6%

Leave-one-signature-out paired transitions

Source	Masked signature	Aligned only	Mask only	Both pass	Difference
C3	bound_relation	50	0	450	-0.100
C3	multi_argument_function	50	0	450	-0.100
C3	relation_arity_3	100	0	400	-0.200
C3	self_composed_function	50	0	450	-0.100
C3	temporally_indexed_recurrence	50	0	450	-0.100
C3	unobserved_dependency	100	0	400	-0.200
C3	unobserved_selector	50	0	450	-0.100
C_rand	bound_relation	33	0	32	-0.066
C_rand	multi_argument_function	12	0	53	-0.024
C_rand	relation_arity_3	13	0	52	-0.026
C_rand	self_composed_function	6	0	59	-0.012
C_rand	unobserved_selector	1	0	64	-0.002
DeepSeek_grammar	multi_argument_function	15	0	0	-0.156

Motif-disabled cumulative candidate attrition

Source	Stage	Passed	Denominator	Rate
C3	J0	1500	1500	100.0%
C3	J1	1168	1500	77.9%
C3	J2	1168	1500	77.9%
C3	J3	0	1500	0.0%
C3	J4	0	1500	0.0%
C3	J5	0	1500	0.0%
C_rand	J0	1500	1500	100.0%
C_rand	J1	1040	1500	69.3%
C_rand	J2	1040	1500	69.3%
C_rand	J3	0	1500	0.0%
C_rand	J4	0	1500	0.0%
C_rand	J5	0	1500	0.0%
DeepSeek_grammar	J0	288	288	100.0%
DeepSeek_grammar	J1	236	288	81.9%
DeepSeek_grammar	J2	236	288	81.9%
DeepSeek_grammar	J3	0	288	0.0%
DeepSeek_grammar	J4	0	288	0.0%
DeepSeek_grammar	J5	0	288	0.0%